

SENNA FOODS

Logistics Flow Implementation

Implementation Engineer Intern Assignment

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Target Role: Implementation Engineer Intern

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System Scope: SAP ERP ↔ LMS ↔ Transporter Portal ↔ Driver App Integration

Format: Comprehensive Architecture & Engineering Solution

ASSIGNMENT SCOPE & SOURCE OF TRUTH

This submission contains the complete system design, data mappings, API/Webhook specifications, failure handling strategies, and Python validation script for the Senna Foods Logistics Flow. All data mappings and validation tests strictly adhere to the official assignment specification.

1. Executive & Business Overview

Senna Foods manufactures and distributes packaged food products across India. Billed orders are generated inside **SAP ERP**. Historically, logistics operations relied on manual Excel spreadsheets and informal WhatsApp updates, leading to data mismatches and lack of visibility regarding real-time order delivery vs. returns.

Target Logistics Workflow:

SAP ERP → LMS Engine → Transport Company Portal → Driver Partner App → LMS Engine → SAP ERP

The Logistics Management System (LMS) serves as the core operational middleware. SAP remains the primary **System of Record**; the LMS reads master data and returns item-level delivery results without directly mutating SAP master tables.

System / Component	Primary Role & Function	Integration Protocol
SAP ERP	Central System of Record for billing, products, and customer orders.	Current: Daily CSV File Target: REST API (JSON)
LMS Engine	Trip planning, route optimization, validation, and execution tracking.	REST API / Webhook Engine
Transporter Portal	Web console for logistics partners to accept trips and assign trucks/drivers.	Web Portal / REST API
Driver Partner App	Offline-first mobile app for drivers to execute stops, capture POD photos & GPS.	HTTPS Webhooks (X-Signature)

2. Task 1 — Data Mapping & Integration Hand-Offs

System Hand-off Specification Table

#	Data Description	From	To	Method	Validation & Failure Handling
1	Daily Deliveries	SAP	LMS	Current: Daily CSV Target: REST API	LMS validates master data. Invalid rows (missing weight, unknown shop) are rejected to Error Quarantine.
2	Planned Trip Details	LMS	Transporter	REST API / Web Portal	Transporter cannot accept trip until truck capacity and zone constraints pass.
3	Driver Assignment	Transporter	LMS	Portal Form Submission	Trip remains 'Sent to Transporter'. Driver app cannot fetch trip until assigned.
4	Mobile Trip Sync	LMS	Driver App	REST API (JSON)	App caches trip locally for offline execution in dead zones.
5	Stop Execution Result	Driver App	LMS	HTTPS POST Webhook	Secured with X-Signature. Saves locally when offline (recorded_at) and syncs online (sent_at).
6	Delivery Results	LMS	SAP	REST API (JSON)	Item-level posting. Retries with backoff if SAP is temporarily unreachable.

SAP to LMS Field Mapping Table (Including Gross Weight BRGEW)

SAP Field	LMS Target Field	Type	Req?	Transformation & Technical Rule
VBELN	delivery_number	text(10)	Yes	Pads/preserves leading zeros to 10 chars (e.g. 0080001234).
POSNR	line_number	integer	Yes	Strips leading zeros to integer (e.g. 10, 20).
KUNNR	shop_code	text	Yes	Strips leading zeros (e.g. 0000100317 → 100317). Looks up Shop Master.
NAME1	shop_name	text	Yes	Direct customer name mapping (e.g. Ganesh Mart).
ORT01	city	text	Yes	City location for shop master.
PSTLZ	pin_code	text(6)	Yes	Postal code used for LMS delivery zone mapping.
MATNR	product_code	text	Yes	Material SKU code (e.g. FG-3010).
ARKTX	product_name	text	No	Additional product description string.
LFIMG	quantity	integer	Yes	Billed order item quantity.
VRKME	unit	text	Yes	Unit Conversion: Converts CS → CASE; PC → PIECE.
BRGEW	delivery_item.weight_kg	decimal	Yes	Gross Weight: Line item weight. Missing BRGEW blocks/quarantines delivery.

MISSING GROSS WEIGHT (BRGEW) FAILURE HANDLING

In the sample dataset, Delivery 0080001236 has a missing BRGEW value. When BRGEW is missing, LMS validation immediately flags the record. The delivery **does not silently proceed to trip creation** because total truck payload capacity cannot be computed. The delivery is quarantined in the Error Queue until corrected in SAP.

System Flow & Validation Architecture Diagram

[Diagram Image: diagram/logistics_flow.png]

3. Task 2 — Send Trip Results to SAP

Trip Status Lifecycle Flow

Status	Next Status	Triggering Event / Action	Actor / System
Planned	Sent to Transporter	LMS routing engine finishes optimization and dispatches trip to portal.	LMS Engine (Auto)
Sent to Transporter	Assigned	Transporter accepts trip and assigns truck registration and driver details.	Transporter Dispatcher
Assigned	Started	Driver taps 'Start Trip' on mobile app at warehouse gate.	Driver Partner App
Started	Completed	Driver finishes all assigned shop stops and uploads final POD.	Driver Partner App
Completed	<i>Final State</i>	LMS posts item-level results back to SAP ERP.	LMS Engine → SAP

Timestamp Distinction & Result Granularity

- **recorded_at vs. sent_at:** recorded_at is the exact device timestamp when the driver tapped 'Save Result' locally on the phone (works completely offline in basements). sent_at is the timestamp when the app uploaded the payload to LMS over cellular network. Using recorded_at guarantees precise SLA measurement.
- **Per Item Result Choice:** Delivery results are reported **PER ITEM** to SAP because billing and stock ledgers are maintained per SKU line. Item-level reporting enables SAP to issue partial credit notes specifically for returned/damaged items while billing delivered items.

Sample Part-Delivered Webhook Payload (JSON)

```
{ "event_id": "evt_01J9ZQ4K7M2X", "event_type": "stop.result_saved", "sent_at":  
"2026-10-02T11:42:15+05:30", "trip_id": "TRP-BLR-20261002-07", "truck_number": "KA-01-AB-1234",  
"driver": { "name": "Ravi K", "phone": "+91-98450-00000" }, "stop": { "sequence": 2,  
"sap_delivery_number": "0080001235", "shop_code": "0000100317", "result": "PART_DELIVERED", "items": [  
{ "material": "FG-1001", "loaded_qty": 10, "delivered_qty": 10, "returned_qty": 0 }, { "material":  
"FG-3010", "loaded_qty": 40, "delivered_qty": 36, "returned_qty": 4, "return_reason": "DAMAGED" } ],  
"pod_photo_url": "https://partner.example.com/pod/TRP-BLR-20261002-07/2.jpg", "recorded_at":  
"2026-10-02T11:40:03+05:30", "location": { "lat": 12.9352, "lng": 77.6245 } } }
```

Six Given / When / Then Test Cases (Including Failure Cases)

#	Scenario	Given (Initial State)	When (Driver Action)	Then (Expected Result)
1	Fully Delivered	Delivery 0080001234 loaded with 24 CS Oats, 12 CS Ketchup.	Driver delivers 100% items and captures POD photo.	LMS updates status to DELIVERED. SAP posts full Goods Issue.
2	Part Delivered	Delivery 0080001235 loaded with 40 CS Noodles.	Driver delivers 36 CS; 4 CS damaged & returned.	LMS sets PART_DELIVERED. SAP posts 36 delivered, 4 returned (DAMAGED).
3	Full Return	Delivery 0080001238 assigned to shop.	Shop closed; shop keeper refuses delivery.	LMS sets RETURNED (SHOP_CLOSED). Delivery flagged for redelivery.
4	Missing Weight (Failure 1)	Delivery 0080001236 received from SAP with missing BRGEW.	LMS validates CSV during daily ingestion.	FAILS: LMS quarantines delivery to Error Queue; alerts SAP team.
5	Unknown Shop (Failure 2)	Delivery 0080001237 received for customer 0000100999.	LMS cross-references shop master DB.	FAILS: LMS rejects delivery to Quarantine; displays summary error log.
6	Duplicate Webhook	Event evt_01J9ZQ4K7M2X already processed in LMS.	Driver app re-sends same event_id on retry.	LMS detects duplicate event_id, skips processing, returns HTTP 200.

4. Task 3 — APIs, Webhooks & System Reliability

Master Data REST APIs Specification

Master Entity	Method	Endpoint	Main Required Fields	Operational Purpose
1. Shop	POST	/api/v1/shops	shop_code, shop_name, city, pin_code	Onboards retail shop customer master records from SAP into LMS.
2. Delivery Location	POST	/api/v1/delivery-locations	shop_code, latitude, longitude, delivery_window	Configures map pins and store receiving hours.
3. Zone	POST	/api/v1/zones	zone_code, zone_name, city, pin_codes	Groups postal codes into delivery zones for routing.
4. Truck Type	POST	/api/v1/truck-types	truck_type_code, max_weight_kg, max_volume_m3	Defines truck capacity constraints for routing engine.
5. Transport Company	POST	/api/v1/transport-companies	transporter_code, name, contact_email, phone	Registers transport partners for portal access.
6. Freight Rate	POST	/api/v1/freight-rates	transporter_code, zone_code, truck_type_code, rate	Sets contract freight pricing per zone and truck.
7. Product	POST	/api/v1/products	product_code, product_name, unit, weight_kg_per_unit	Onboards food SKU master details and unit weights.

Driver Partner App Webhook Events Table

Event Name	Sender System	When Sent	Main Payload Data
1. trip.started	Driver Partner App	Driver taps 'Start Trip' at warehouse gate.	event_id, trip_id, truck_number, driver, start_time
2. stop.result_saved	Driver Partner App	Driver saves delivery result at shop stop.	event_id, trip_id, sap_delivery_number, items[], recorded_at
3. trip.completed	Driver Partner App	Driver completes all assigned stops on trip.	event_id, trip_id, completion_time, total_stops

Technical Failure & Security Controls

- **Security Authentication (X-Signature Header):** The Driver App sends webhooks as HTTPS POST requests containing an X-Signature header. LMS verifies the signature before processing. Invalid signature → reject with HTTP 401 Unauthorized.
- **Duplicate Event Handling (event_id Idempotency):** LMS reads event_id from the body. If event_id exists in logs, LMS skips reprocessing business actions and returns HTTP 200 OK so the app stops retrying.
- **LMS Downtime / Retry Handling:** If LMS is temporarily unavailable, the Driver App caches events locally in device storage and retries transmission with exponential backoff delays. After extended failures, events move to a dead-letter log.
- **Unknown Shop Validation (0000100999):** Ingestion validates KUNNR against LMS Shop Master. If unknown, LMS rejects the delivery to Quarantine Error Queue and alerts SAP team. Unknown shops are **not auto-created**.

5. Bonus — Python Delivery Validation Script

The script `bonus/process_deliveries.py` parses `SAP_delivery_sample.csv`, groups line items by VBELN, computes total weights, validates shop master codes, and highlights validation errors without crashing.

Script Execution Output Log (Representative Output)

```
===== SENNA FOODS LOGISTICS - SAP DELIVERY
SUMMARY REPORT ===== Delivery: 0080001234 Shop:
Sri Balaji Stores Items: 2 Total Weight: 27.60 KG ----- Delivery:
0080001235 Shop: Ganesh Mart Items: 2 Total Weight: 17.20 KG -----
Delivery: 0080001236 Shop: New Fresh Mart Items: 1 Total Weight: INCOMPLETE (Missing weight data)
----- Delivery: 0080001237 Shop: Om Provisions [UNREGISTERED IN
LMS] Items: 1 Total Weight: 5.60 KG ----- ...
===== VALIDATION WARNINGS =====
WARNING: Delivery 0080001236 has missing weight. ===== CRITICAL
ERRORS ===== ERROR: Shop 0000100999 is not present in LMS.
Processing complete.
```

6. Client Email to SAP Team & Conclusion

Non-Technical Email Draft to Senna Foods SAP Team (<150 words)

Subject: Action Required: Data Validation Issues in Daily SAP Delivery Feed

Dear SAP Operations Team,

During our daily LMS data validation, we identified two critical data issues in the latest SAP delivery file:

- Missing Gross Weight (BRGEW) on Delivery 0080001236:** The item weight is missing. Without weight data, our system cannot calculate truck capacity or allocate appropriate vehicles.
- Unregistered Customer/Shop (0000100999 - Om Provisions) on Delivery 0080001237:** This shop code does not exist in the LMS master database. Drivers cannot navigate to or execute deliveries for unmapped stores.

Required Action:

- Please update the gross weight (BRGEW) for Delivery 0080001236 in SAP.
- Please onboard customer 0000100999 into the LMS shop master database via the shop sync interface.

Once updated, please re-trigger the file sync so trip planning can proceed.

Best regards,

Logistics Implementation Team | Senna Foods Logistics Support

Conclusion & Architectural Summary

This implementation design establishes a robust, failure-resilient bridge between SAP ERP and field logistics execution. By enforcing strict master data validation during ingestion, securing mobile webhooks with X-Signature, and preserving item-level return granularities, Senna Foods achieves end-to-end delivery visibility while maintaining SAP as the single System of Record.

Note: Advanced queuing architectures (e.g. RabbitMQ/SQS) and real-time CDC sync pipelines are documented as future production recommendations beyond the scope of this assignment.